Estimating Cuba's 2021–2026 population collapse

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Working paper (preprint) · 30 July 2026 · Not peer-reviewed

One in four Cubans gone in five years (official 2021 base by end-2026)

Monte Carlo interval estimates, an infant-mortality sentinel, and triangulation of independent registers

Abstract

Background. Cuba is losing population at one of the fastest peacetime rates in the Western Hemisphere. Against the official 2021 headcount, cumulative loss by end-2026 approaches one in four people. No census has been completed since 2012, and the national statistics office (ONEI) removed more than one million people from its published headcount in a single 2024 revision. End-2024 estimates still span roughly 8.0–11.0 million. This paper quantifies the 2021–2025 decline, decomposes migration versus internal demographic collapse, and projects to end-2026.

Methods. We solve the demographic accounting identity with Monte Carlo simulation ($N = 10^6$) under four prior sets (Models A–D), with Gaussian-copula dependence among baseline overstatement, migration, and death registration. Preferred Model D tightly bounds death under-registration using an infant-mortality consistency check (elasticity calibrated on the Venezuelan collapse analog), while retaining wide migration uncertainty. External registers are used as ceilings or corroboration, not as circular confirmation of Model D itself. Age-standardized excess mortality is reported separately from crude registered excess.

Results. Across models, cumulative 2021–2025 loss spans roughly **1.8–2.4 million** (structural uncertainty). Preferred Model D places the living population near **8.59 million** at end-2025 (median loss **2.21 million**, **20.5%** on a corrected 2021 base; within-model 90% interval for loss roughly 1.84–2.60 million). Emigration accounts for about **91%** of the absolute loss. Age-adjusted excess deaths in 2024–2025 are about **40,000–60,000**; crude registered excess over 2020–2025 is larger ($\sim$153,000) and must not be conflated with the age-adjusted figure. A nowcast places Model D near **8.29 million** by end-2026 ($\sim$300,000 further loss).

Conclusions. Depopulation is predominantly migratory in the headcount, but is reinforced by collapsing births, elevated mortality, and accelerated ageing. Cumulative loss from 2021 approaches roughly one in four people by end-2026 when measured against the official 2021 base (a different definition from the Model D -20.5% figure for end-2025).

Keywords: Cuba; demography; emigration; natural decrease; infant mortality; fertility; ageing; statistical data quality.

1 Introduction

Counting Cubans has become contested. The last completed census was in 2012; the census planned for 2022 has been repeatedly postponed and is now promised for 2026 [ONEI, 2022]. Without a census, published population depends on projections and a lagging migration register. In July 2024, ONEI reported an "effective population" of 10,055,968 at end-2023 (about one million below prior published levels), then 9,748,007 (end-2024) and 9,434,593 (end-2025) [ONEI, 2024, 2025].

Three incompatible series coexist (Figure 3a): the United Nations World Population Prospects path near 10.9 million (still assuming net emigration of order 2×10^4 per year) [United Nations, Department of Economic and Social Affairs, Population Division, 2024]; ONEI's revised official series (9.43 million at end-2025); and independent estimates placing the living population nearer 8.0–8.6 million [Albizu-Campos Espiñeira, 2023, 2024]. Almost three million people separate the extremes. This paper estimates the 2021–2025 decline with honest intervals, decomposes drivers into migratory and internal components, audits official migration accounting against destination registers, and projects to end-2026.

Central thesis. The headcount loss is mostly emigration, but a second "internal" engine (birth collapse, excess mortality, and ageing) would sustain contraction even if emigration slowed. The two engines are not independent: selective exit of working-age adults and women of reproductive age causes much of the fertility collapse, so many "births that do not occur" are a delayed migratory effect.

2 Methods

2.1 Accounting identity and Monte Carlo

Registered ONEI aggregates over 2022–2025 are fixed anchors [ONEI, 2025]: births $B^{\text{reg}} = 325{,}233$, deaths $D^{\text{reg}} = 502{,}149$, and identity-consistent net emigration $R = 1{,}501{,}706$. Official end-2021 population is $P^{\text{off}}_{2021} = 11{,}113{,}215$. Each Monte Carlo draw samples latent corrections and applies

$$
\begin{aligned}
P_{2021} &= P^{\text{off}}_{2021} - U_0, &(1)\\
D &= D^{\text{reg}}\, D_{\text{fac}} \ \ (+\text{ optional epidemic pulse in B/C}), &(2)\\
M_{\text{net}} &= R\, M, &(3)\\
\Delta &= (D - B) + M_{\text{net}}, &(4)\\
P_{2025} &= P_{2021} - \Delta, &(5)
\end{aligned}
$$

where U_0 is end-2021 baseline overstatement (people already effectively abroad but still on the roll), $M \geq 1$ scales ONEI net emigration, and $D_{\text{fac}} \geq 1$ scales registered deaths. Percent loss uses the *corrected* base P_{2021}, not P^{off}_{2021}. We draw $N = 10^6$ scenarios. In Model D, (U_0, M, D_{fac}) are coupled by a Gaussian copula on latent normals with correlation matrix

$$
\rho = \begin{pmatrix} 1 & 0.55 & 0.30 \\ 0.55 & 1 & 0.25 \\ 0.30 & 0.25 & 1 \end{pmatrix}, \tag{6}
$$

then mapped to Beta margins (Table 1). Within-model 90% intervals are sampling uncertainty under a fixed prior set. The **range across Models A–C/D** ($\sim$1.8–2.4 million of cumulative loss) is structural uncertainty and must not be read as a single CI [Pérez-Riverol, 2026c,b].

2.2 Four models and preferred priors

- **A (conservative):** modest U_0 and M near ONEI; small death multiplier (implementation: `model_from_2021.py`).
- **B (crisis-adjusted):** higher M, death under-registration up to $\sim$14%, plus an explicit epidemic pulse.

- **C (worst case):** still higher U_0/M and under-registration (cap $\sim$18%) with a large elderly mortality pulse.
- **D (preferred, sentinel):** migration priors comparable to B; death under-registration *tightened* after the infant-mortality consistency check (Table 1).

Model D is a preferred central path inside the across-model envelope, not a uniquely identified point estimate. Models B and D yield nearly identical headcount medians because migration priors dominate absolute loss; they differ mainly in how mortality is justified.

Table 1: Model D prior specification (margins after Gaussian copula). Beta scales are written so that $U_0 \in [0, 700{,}000]$, $M \in [1, 1.72]$, and $D_{\text{fac}} \in [1, 1.09]$. Birth noise is independent $B = B^{\text{reg}}\mathcal{N}(1, 0.01^2)$.

Symbol	Margin	Interpretation
U_0	$700{,}000 \times \mathrm{Beta}(2.0, 2.4)$	End-2021 roll overstatement
M	$1 + 0.72 \times \mathrm{Beta}(2.2, 2.4)$	Multiplier on ONEI net emigration R
D_{fac}	$1 + 0.09 \times \mathrm{Beta}(1.8, 3.2)$	Residual death under-registration (at-home elderly channel)

2.3 Infant-mortality consistency check

Infant mortality rose from 5.0 to 9.9 per thousand live births (2019–2025), a $\sim$98% increase ($\sim$39% year-on-year in 2024–2025). Infant deaths are comparatively hard to hide (hospital occurrence; international scrutiny), so IMR is treated as a high-integrity sentinel of health-system stress, not as a direct estimator of total deaths.

As a *calibration analogy*, Venezuela's collapse saw IMR rise $\sim$76% (roughly 2012–2017) alongside an all-cause crude death-rate rise of order $\sim$25%, implying an elasticity

$$e = \frac{\mathrm{d}\ln(\mathrm{CDR})}{\mathrm{d}\ln(\mathrm{IMR})} \approx 0.30 \tag{7}$$

(working range 0.22–0.40 used only for sensitivity) [Calibration note, 2017]. Applied to Cuba's IMR rise, $(1 + 0.98)^{0.30} - 1 \approx 30\%$ predicted all-cause CDR increase. Registered Cuban CDR rose from 9.7 to 12.9 per thousand by 2024 ($\sim$33%), inside that band [ONEI, 2025]. This is a **consistency check**, not structural identification: it argues against very large *hidden institutional* death undercounts and motivates bounding residual under-registration in Model D to the at-home elderly channel during funeral-system stress (central $\sim$3%, cap 9%; Table 1) [Pérez-Riverol, 2026b].

2.4 Excess mortality

Two distinct quantities are reported and must not be added or substituted.

Crude registered excess. On crisis-adjusted mid-year stocks $\tilde{P}_t$, define a no-crisis counterfactual CDR with a mild ageing drift from the 2019 level 9.7 per thousand, $\mathrm{CDR}_t^{\mathrm{cf}} = 9.7 + 0.16(t - 2019)$. Then

$$E_t^{\text{crude}} = D_t^{\text{reg}} - \mathrm{CDR}_t^{\mathrm{cf}} \cdot \tilde{P}_t/1000. \tag{8}$$

Summing 2020–2025 yields $\sum E_t^{\text{crude}} \approx 153{,}000$ (of which $\sim$66,000 in 2024–2025). Most of this toll is already inside registered death totals.

Age-schedule excess (primary scientific claim).

To ask how many deaths exceed what the *already aged* population would produce under 2019 age-specific rates $m_{a,2019}$,

$$E_t^{\text{age}} = D_t^{\text{reg}} - \sum_a m_{a,2019}\, P_{a,t}, \tag{9}$$

using ONEI age structure updated for recent ageing (Kitagawa-style rate/composition separation in spirit; full cohort reconstruction is not claimed) [Kitagawa, 1955]. For 2024–2025 this yields about **40,000–60,000** excess deaths; the interval reflects uncertainty in age stocks (official versus Model D-consistent populations) and in the 2019 schedule. Illustrative point comparison in Figure 6b: expected $\sim$112.5k and $\sim$113.8k versus registered 128.1k and 136.2k. Mortality rates at ages 60+ remaining 11–27% above 2019 support elevated age-specific risks, not composition alone.

2.5 External registers (non-circular)

Published levels are grouped by information content (Figure 4):

- **Ceilings** (do not purge emigrants): UN World Population Prospects, MINSAP denominator, electoral roll.
- **Official:** ONEI revised headcount (9.43 M at end-2025).
- **Exit-correcting / independent:** housing$\times$occupancy ($\sim$8.7 M), Albizu-Campos 2023/2024 (8.0–8.6 M).

Model D (8.59 M) is the study estimand and is *not* counted as external validation of itself. Agreement is assessed against the exit-correcting family; ceilings bound the upside. The probable band from those external sources is near 8.0–8.9 million.

2.6 Destinations ledger and migration bridge

Settled migrants only (avoiding transit double-counting): United States $\sim$800,000; Spain $\sim$130–135,000; Uruguay $\sim$35,000; other $\sim$255,000; floor $\sim$1.05 million [Pérez-Riverol, 2026a]. Encounter counts and nationality applications are separate concepts and are never added into the settled floor (Table 2).

Table 2: Bridge from settled destination floor to Model D net emigration (2022–2025). Concepts are mutually exclusive unless labeled.

Concept	Approx. count	Role
Settled destinations floor	1.05 M	Lower bound (unique settled persons)
+ U.S. undercount, limbo, maritime gaps	$\rightarrow\sim$1.5 M	Toward ONEI revised net
ONEI identity-consistent net R (2022–25)	1.50 M	Official flow after revision
Albizu-Campos–type net	$\sim$1.8 M	Independent upper narrative
Model D median $R \times M$	$\sim$2.0 M	Preferred migratory engine
U.S. encounters 2022–2024 (not settled)	$\sim$0.85 M	Events; do not add to floor
Spain nationality applications	$\sim$0.30 M	Many applicants still in Cuba

If true net emigration were nearer 1.5 than 2.0 million, living population would sit toward $\sim$8.9 rather than $\sim$8.6 million; cumulative loss would remain of order one quarter and emigration would still dominate absolute loss.

2.7 End-2026 nowcast (Model D)

Conditional on end-2025 draws, 2026 uses births centered at 65,682 ($\mathcal{N}$ noise 3%), deaths centered at $1.04 \times$ 136,214 times the same D_{fac}, and a three-regime net emigration mixture with probabilities $(0.35, 0.35, 0.30)$ on centers $(150{,}000,\ 210{,}000,\ 300{,}000)$ plus 15% log-scale noise. The $\sim$4% death lift is a scenario assumption under continued sentinel stress, not a forecast identity. Paths to $\sim$7 million by 2030 in Figure 3a are **illustrative trend scenarios**, not probabilistic forecasts.

3 Results

3.1 Headline estimates

Table 3 summarizes end-2025 medians. The scientifically primary uncertainty statement is the across-model loss envelope ($\sim$1.8–2.4 million). Preferred Model D yields median population **8.59 million** and loss **2.21 million** (20.5% on the corrected base; within-model 90% interval for loss 1.84–2.60 million). Migration's share of absolute loss is about **91%**. Models B and D converge near 8.6 million because their migration multipliers are similar; Model D mainly replaces B's wider mortality pulse with the sentinel-bounded D_{fac}. Percentage language depends on the base: against official 2021 (11.11 M) end-2025 loss is roughly 20–23%; against the corrected study base it is about 16–23% across models. A 2026 nowcast under the mixture in Section 2.7 places Model D near **8.29 million** (further loss $\sim$300,000; across-model band roughly 7.9–8.9 million).

Table 3: Models A–E: population loss 2021–2025 and end-2025 population (medians). Model E is provisional.

Model	Main assumption	Loss (M)	%	Pop. end-2025 (M)
A · conservative	ONEI-anchored	1.80	16.4	9.16
B · crisis-adjusted	under-reg. + independent migration	2.19	20.4	8.57
C · worst case	analog upper bound	2.41	22.6	8.26
D · sentinel (preferred)	IMR sentinel + triangulation	2.21	20.5	8.59
E · vital reconstr. (prov.)	ageing + HSDS $\rightarrow B^*, D^*$	2.13	19.2	8.96

3.2 Model E: vital reconstruction (provisional)

As a companion path, Model E reconstructs *true* deaths and births from a yearly health-system deterioration score S_t built from ageing, health-system decadence, and documented official underreporting (including COVID cause-attribution misuse as an integrity feature, not as a $\times N$ multiplier on all deaths). Ageing enters an expected-death schedule; decadence raises an uplift $f(S_t)$; births stay near the register with crisis fertility and selective exit. Closing the identity $P^* = P^*_{\text{start}} + \sum(B^* - D^* - M^*)$ with a Model D-like migration bridge yields median end-2025 population about **8.96 million** (loss 2.13 million, 19.2% on the corrected base; migration share $\sim$88%). Figure 1 shows S_t and reconstructed versus registered vitals; Figure 2 places E beside A–D. Preferred public claims remain Model D until an adversarial gate promotes E.

3.3 Triangulation of the living population

Sources that do not purge emigrants (electoral roll, MINSAP health denominator, and related registers) sit high and function as ceilings. Sources that correct for exit (housing occupancy $\sim$8.7 M; independent estimates 8.0–8.6 M) converge near Model D (8.59 M) without treating Model D as an external datum. The

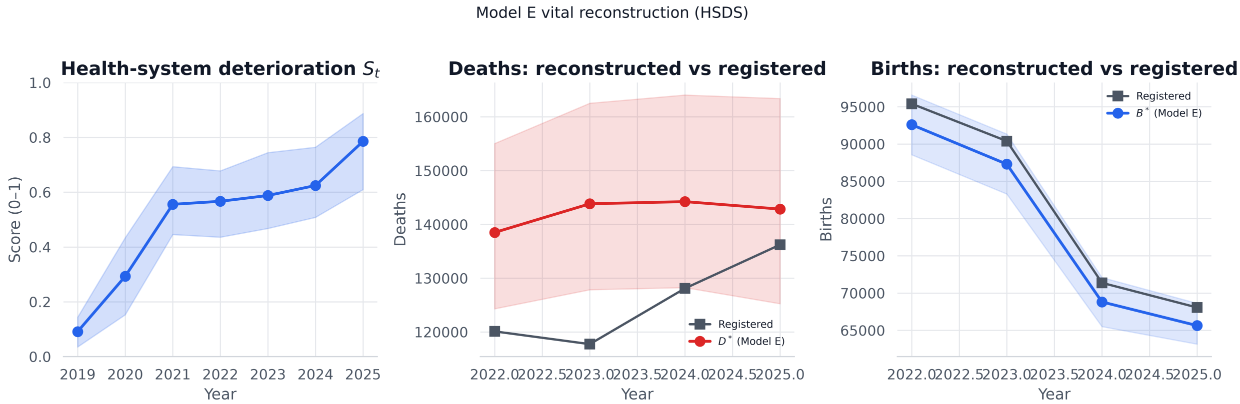


Figure 1: Health-system deterioration score S_t and Model E reconstructed deaths/births versus registered series (2022–2025).

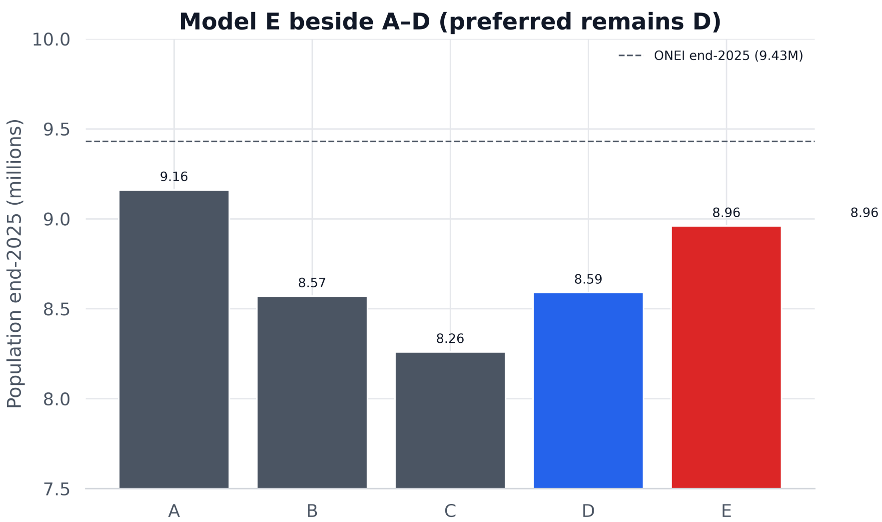
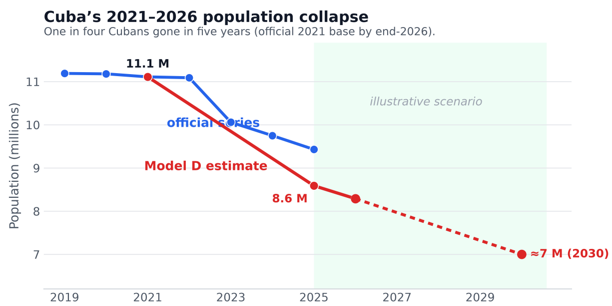
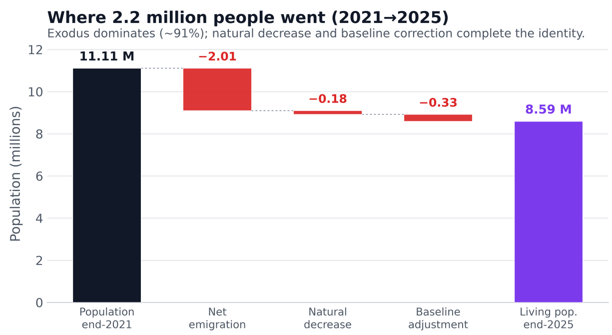


Figure 2: End-2025 population medians for Models A–E. Preferred public model remains D (blue); E (red) is provisional.



(a) Official versus estimated living population, 2021–2030 (Model D path in red; post-2026 segment is an illustrative scenario).



(b) Accounting decomposition of the 2021–2025 loss (emigration dominates).

Figure 3: Headline population path and accounting decomposition.

official 9.43 M sits between the two families, closer to inflated registers: a ceiling, not a floor (Figure 4). Between 2018 and 2023, actual voters fell 16.7% (1.24 million fewer), far more than the roll itself (−5.9%). That pattern is *compatible with* physical absence but also with abstention, logistics, and political factors; it is not used as a unique person count.

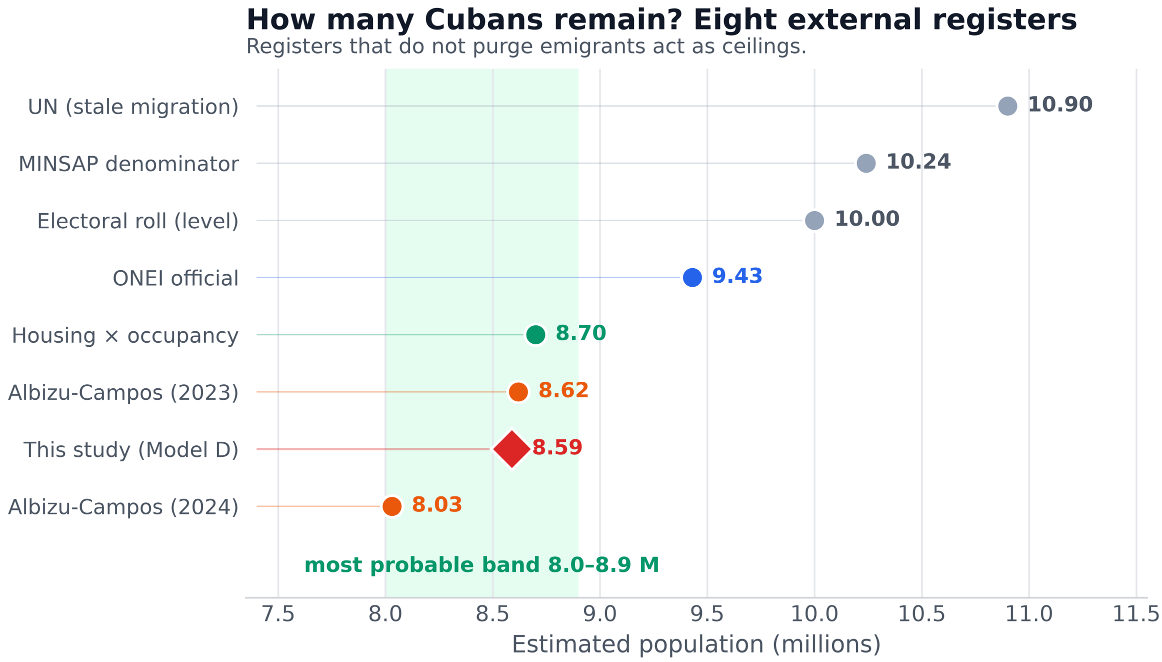


Figure 4: Triangulation of end-2025 population levels across independent sources.

3.4 Destination confirmation of the migratory floor

Table 2 separates settled counts from encounters and nationality applications. The settled floor (∼1.05 M) confirms that emigration is large while remaining a lower bound: U.S. undercount, limbo statuses, and maritime routes are incompletely captured. Closing those gaps moves the implied net toward ONEI's ∼1.5 million; more aggressive estimates (Albizu-Campos ∼1.8 M; Model D median ∼2.0 M) sit at the upper edge. If true net emigration is nearer 1.5 than 2.0 million, living population sits toward the high end of our band (∼8.8 rather than 8.5 M). In all cases, cumulative loss remains of order one quarter and emigration remains ∼91% of the absolute Model D loss.

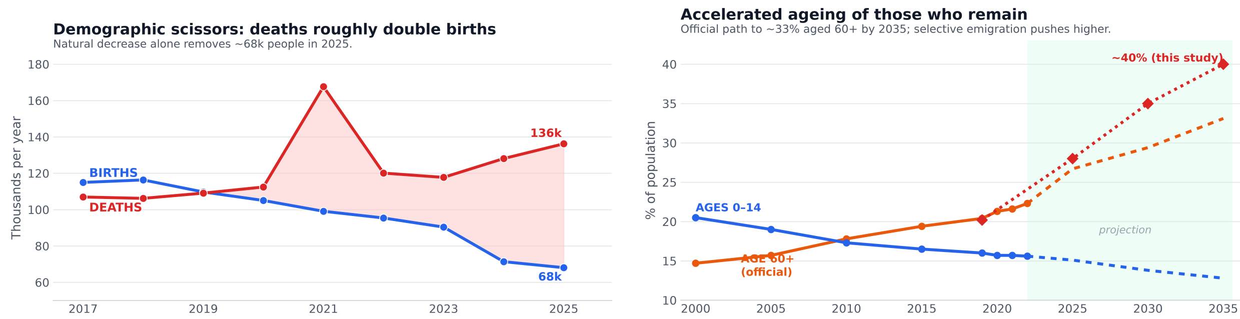
3.5 Internal demographic engine

Reducing the crisis to emigration alone is incomplete. Emigration can reverse; missing births and an aged residual population cannot. Useful distinction: **realized loss** (exit-dominated) versus **future inertia** (natural decrease increasingly important).

Deaths nearly double births. In 2019 the natural balance was still slightly positive (+636); in 2025 deaths exceeded births by **68,149** (68,064 births versus 136,214 deaths; Figure 5a). That 2025 deficit matches the 2021 COVID peak without a pandemic: natural decrease is now structural.

Fertility. The total fertility rate fell to 1.29 children per woman in 2025, the lowest since 1958 and well below replacement (2.1). Two intertwined causes: emigration removes young women (the flow is majority female), and those who remain defer or forgo births under blackouts and scarcity.

Ageing. Official projections move the share aged 60+ from 20.4% (2019) to 26.7% (2025) and toward 33% by 2035, but they assume a larger living stock than Model D. Under a smaller, youth-depleted stock the elderly share is higher still (order ∼29% already in 2025 and, approximately, ∼40% by 2035 as a first-order sketch requiring a full cohort projection). Median age reached **45** years in 2025 (Figure 5b).



(a) Births versus deaths (“scissors”): deaths roughly double births by 2025.

(b) Accelerated ageing as working-age adults leave.

Figure 5: Internal demographic engine: natural decrease and accelerated ageing.

Life expectancy. Official life expectancy at birth peaked in 2011–13 (78.5 years) and fell to 77.7 in 2018–20, before the worst of the crisis (Figure 6a). An independent 2021 estimate places it near 71 years [Albizu-Campos Espiñeira, 2023, Brønnum-Hansen and Albizu-Campos Espiñeira, 2023]. Life expectancy re-expresses the same mortality shock rather than providing independent corroboration; its magnitude (about 6.5 years in a single year in the independent path) is exceptional for peacetime. Translated to years of life lost, crude excess of ∼153,000 deaths in 2020–2025 implies more than one million person-years lost, concentrated among the elderly.

Infant mortality and food insecurity as mechanisms. Infant deaths rose from 461 (2018) to 715 (2022); the rate reached 9.9 per thousand in 2025 (+98% since 2019). Cuba imports 70–80% of food consumed; agricultural GDP fell 61.7% between 2018 and 2024; WFP estimates more than four million Cubans in food insecurity (∼22.5% anaemia in early childhood). These facts do not yield a malnutrition death series (none exists); they are treated only as plausible amplifying pathways for infant and elderly mortality and for depressed fertility.

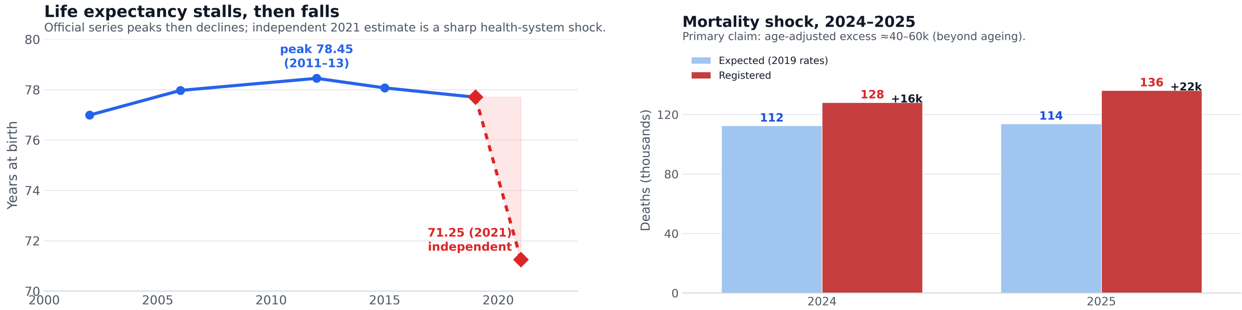
3.6 Age-adjusted excess in 2024–2025

Infant mortality remains a high-integrity early-warning sentinel because infant deaths occur in hospitals and are hard to hide. Crude registered excess versus the drifted no-crisis CDR is ∼153,000 over 2020–2025 (peak 2021). After removing compositional ageing via the 2019 age-schedule counterfactual, 2024–2025 still show about **40,000–60,000** excess deaths (Figure 6b). Mortality rates at ages 60+ remain 11–27% above 2019: mortality rises not only because there are more elderly people, but because age-specific rates themselves are elevated.

3.7 Who leaves and who remains

About **77%** of emigrants are aged 15–59; nearly half are under 35 (Figure 7a). Among those remaining, about 26.7% are 60+ (median age ∼45). Exit hollows the centre of the pyramid, ages the island, and removes those who would have children.

The flow is majority female (∼57%); among reproductive ages roughly 133 women leave per 100 men. Births fell 38% (from ∼110,000 in 2019 to 68,000 in 2025); about half of that fall is fewer women of fertile age (many emigrated) and half is lower fertility per woman (TFR from ∼1.6 to 1.29). The migratory half is hardest to reverse: fertility recovery without mothers does not restore births.

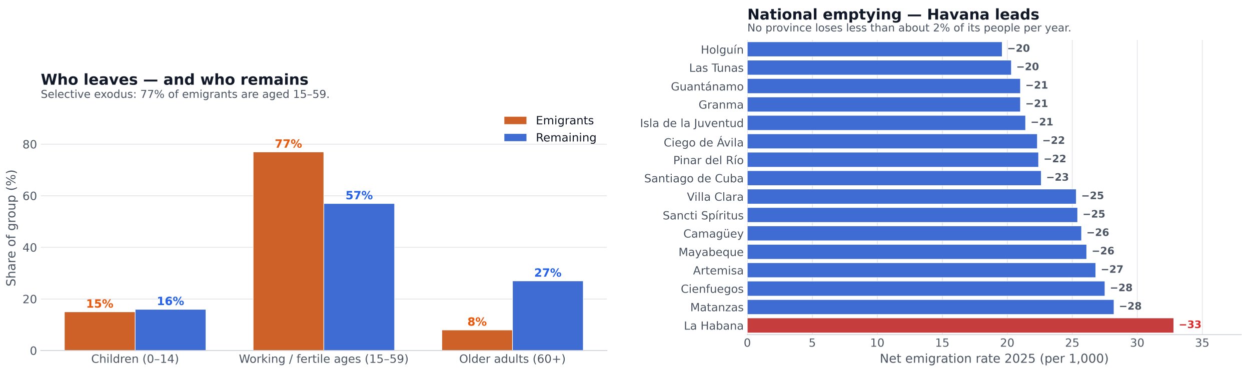


(a) Life expectancy at birth: official triennial series and independent 2021 estimate.

(b) Registered deaths versus an aged 2019-schedule expectation. Primary claim: age-adjusted excess ≈40–60 thousand in 2024–2025. Crude registered excess ∼153,000 over 2020–2025 is a separate quantity.

Figure 6: Mortality signal: life expectancy and age-adjusted excess deaths.

Emptying is national but unequal (Figure 7b). Havana leads at net emigration of −**33 per 1,000** in 2025; western and central provinces empty faster than the east, yet no province loses less than about 2% of its people per year.



(a) Age composition of emigrants versus remaining residents.

(b) Net emigration rate by province, 2025.

Figure 7: Selectivity of exit: age structure and provincial intensity.

3.8 Comparative context and end-2026 path

Historically, contractions of this magnitude appear in deep economic collapse or war (Table 4). Cuba's worst-case annual rate (up to ∼3.9%) matches the Venezuelan peak but sustained longer. Regionally, Latin America and the Caribbean still grow (∼+0.7% per year); Cuba empties at about −3.3% annually (Table 5).

By **end-2026**, the Model D nowcast implies about 300,000 further loss during the year: living population near **8.29 million** (illustrative across-model band 7.9–8.9 M). Against the official 2021 base, cumulative loss approaches ∼2.8 million: roughly **one in four** people. Illustrative trend paths point toward ∼7 million by 2030 (Figure 3a), and near 5.9 million in the worst quartile of exploratory scenarios.

4 Discussion

Model D is preferred because the IMR–CDR consistency check shrinks the need for large hidden-death multipliers and focuses residual uncertainty on migration and on exit-correcting external registers. Model E (vital reconstruction via S_t) lands in the same loss band (∼2.1 M; pop. ∼8.96 M) and raises the natural-

Table 4: Cuba's loss versus selected historical demographic collapses.

Case	Cum. loss	Annual rate	Dominant engine
Syria 2011–20 (war)	$\sim$50%	$>$10%	displacement + deaths
Venezuela 2013–24	$\sim$25%	1–3%	emigration
Zimbabwe 1998–08	$\sim$25%	$\sim$2.5%	emigration
Cuba 2021–25 (Model C)	$\sim$23%	$\sim$5.6%	emigration + internal

Table 5: Annual population growth: Cuba versus the region (approximate recent rates).

Country / region	Annual growth
Guatemala	$+1.7\%$
Bolivia	$+1.4\%$
Dominican Republic	$+1.0\%$
Mexico	$+0.8\%$
Colombia	$+0.7\%$
Latin America & Caribbean	$+0.7\%$
Brazil	$+0.4\%$
Chile	$+0.3\%$
Uruguay	$+0.1\%$
Venezuela	$\approx 0\%$ (after $\sim$25% loss 2013–20)
Cuba	-3.3%

decrease share only modestly ($\sim$12% versus $\sim$9% under D), so it does not yet displace D for public claims. Sensitivity concerns remain first-order: absolute loss is $\sim$91% migratory; migration corrections can share a common outdated-register defect; and within-model intervals look falsely precise relative to the across-model envelope.

Percent language. Against the official 2021 base (11.11 M), cumulative loss by end-2026 approaches $\sim$25% ("about one in four"). Against the corrected Model D base, end-2025 loss is $\sim$20.5% ("about one in five"). These are not interchangeable headlines.

Selectivity couples the engines. Exit is age- and sex-selective; it ages the island, structurally depresses fertility, and (to the extent diaspora composition differs from the resident stock) can shift the composition of those who remain toward groups with worse documented health indicators. The natural balance, positive in 2019, fell to $-68{,}149$ in 2025; age-adjusted excess of 40,000–60,000 in 2024–2025 shows elevated mortality beyond ageing alone. Migration and internal collapse are, in that sense, one coupled process.

4.1 Note on official migration accounting

ONEI's declared 2022 net migration (+991) is incompatible with 313,506 Cuban arrivals to the United States alone that year; the accounting relocates the exodus into a single 2023 jump. Even after that revision, 2024–2025 net balances remain only slightly above U.S.-only recorded entries, while the 2025 balance admits 317,320 gross external emigrations (net $-245{,}264$): continued understatement of exit. This paper therefore treats the official headcount as a ceiling and contrasts it with exit-correcting sources (Figure 4).

5 Limitations

No post-2012 census; ONEI's $\sim$24-month emigration rule lags recent exits; destination microdata are incomplete; the IMR elasticity is a calibration analogy (Venezuela), not a Cuban structural parameter; housing-occupancy assumptions are approximate; ageing under Model D is a first-order sketch pending a full cohort model; life expectancy re-expresses mortality rather than independently corroborating it; 2026–2030 paths are scenarios, not forecasts. Migration corrections share a common outdated-register defect and may partially overlap; destination encounter ceilings can inflate unique persons; single-model intervals give false precision relative to structural uncertainty. External comparison uses ceilings and exit-correcting registers, but no universal administrative register publishes a fully purged emigrant total as a definitive anchor. Food-insecurity and anaemia indicators are treated as plausible amplifying pathways, not as identified causes of the death or birth series.

6 Conclusions

Preferred Model D places Cuba's living population near 8.59 million at end-2025 (not the official 9.43 million), with median loss 2.21 million (20.5% on the corrected base) and emigration about 91% of absolute loss. By end-2026 the nowcast is near 8.29 million; cumulative loss against the official 2021 base approaches one in four. Age-adjusted excess deaths of about 40,000–60,000 in 2024–2025 must be kept distinct from crude registered excess of $\sim$153,000 over 2020–2025. A 2026 census conducted with integrity would be decisive; the working hypothesis is a figure much closer to 8.6 million than to the 10.9 million still printed by the United Nations.

7 Data and code availability

ONEI tables and a machine-readable claim sheet are in the public repository `https://github.com/ypriverol/cubascience` under `demographics/`. Interactive companion: `https://ypriverol.github.io/cubascience/demographics/`. Monte Carlo scripts (`scripts/model_*.py`) write summaries to `artifacts/`. Third-party literature is cited, not redistributed.

Competing interests

None declared.

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